aimshalf

User Guide

Every feature of aimshalf — DFT-1/2 for FHI-aims — and how to use it

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Installation: see the separate Installation Guide

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1. What aimshalf does

Semilocal functionals (LDA, GGA/PBE) underestimate band gaps, mostly because of the electron self-interaction error: the occupied states sit too high. DFT-1/2 (Ferreira, Marques & Teles, 2008) corrects this with Slater's half-occupation idea. For the atom that hosts the hole at the valence band maximum (VBM), it computes the potential created by removing half an electron, the **self-energy potential** (SEP):

$$\mathbf{V_S(r) = [V_{ion}(r) - V_{atom}(r)] \cdot \Theta(r)}, \quad \Theta(r) = A\,[1 - (r/r_{cut})^8]^3 \text{ for } r < r_{cut}, \text{ 0 beyond.}$$

V_S is attractive. Added around every atom of that species, it lowers the valence band and opens the gap, at the cost of a plain PBE calculation. The cutoff radius r_{cut} is not fitted to experiment: it is chosen *variationally*, as the value that maximizes the band gap.

aimshalf automates the whole procedure for the all-electron code FHI-aims, following minushalf (the GMSN/ITA tool for VASP) feature by feature:



- **Valence corrections** remove charge from the orbital(s) at the VBM (attractive SEP).
- **Conduction corrections** (optional) use the orbital(s) at the conduction band minimum (CBM) with a repulsive SEP (amplitude $-|A|$) that raises the conduction band.
- Each corrected orbital (*channel*) gets its own atomic calculation and its own r_{cut}, optimized one after another, as in minushalf.

2. Quick start

Starting from a folder with a converged plain-PBE setup:

1. Put `control.in` (with species defaults and `output band` lines) and `geometry.in` in a folder.
2. Create `aimshalf.yaml` with at least the aims command:

```
aims_cmd: "mpirun -np 8 /path/to/aims.261002.scalapack.mpi.x"   # aims with the aimshalf hooks
```

3. Activate the environment and start:

```
source ~/software/aims_half/.venv/bin/activate
aimshalf execute              # goes to the background; the terminal can be closed
```

4. Follow the run and read the result:

```
cat aimshalf_status.txt       # RUNNING / DONE / FAILED, best so far
tail -f aimshalf_execute.log  # live log (Ctrl-C only stops watching)
```

With no other settings, aimshalf runs a PBE reference, corrects the dominant VBM orbital (code `v`), maximizes the smallest direct gap and reports the optimal r_{cut} and the corrected gap in `aimshalf_status.txt`.

3. Preparing a calculation

3.1 control.in

- A normal FHI-aims input for your system: functional, relativity, k-grid, SCF settings and the species defaults appended.

- **xc pbe** (or the functional that matches `xc` in aimshalf.yaml; `pb` = PBE for the atomic calculations). aimshalf warns if they differ.
- **output band** lines along a band path. aimshalf reads the gap and the band-edge characters on that path. Include the points where the VBM and CBM can be (Γ, X, L for zincblende).
- Spin–orbit coupling (`include_spin_orbit`) is supported; the SOC bands are used.
- Do **not** add `aimshalf_potential` lines yourself: aimshalf writes them in each run folder.

```
xc                 pbe
relativistic       atomic_zora scalar
k_grid             8 8 8
output band  0.0 0.0 0.0   0.5 0.0 0.5   21  G X
output band  0.0 0.0 0.0   0.5 0.5 0.5   21  G L
# ... species defaults (tight recommended for production) ...
```

3.2 geometry.in

Use the plain-DFT (or experimental) structure. Do not relax with DFT-1/2: total energies and forces with the SEP are not meaningful. aimshalf also reads geometry.in to count neighbours of the same species (section 6.4).

3.3 aimshalf.yaml

All settings are optional except, in practice, `aims_cmd`. The file `examples/aimshalf.yaml` lists every key with comments; the full reference is in section 10. A typical file:

```
aims_cmd: "mpirun -np 8 /path/to/aims.261002.scalapack.mpi.x"
correction_code: v          # see section 6
indirect: false             # smallest direct gap (minushalf default)
cut_min: 1.5                # r_cut search range, bohr
cut_max: 5.0
```

> **Note.** Command-line options of `execute` (e.g. `--cut-min`) override the same key in aimshalf.yaml.

4. Commands

Every command prints its options with `aimshalf <command> -h`.

Command	Purpose
`execute`	Full DFT-1/2 correction: choose orbitals, optimize r_{cut}, report the gap
`gen-sep`	Write one self-energy potential file (for inspection or manual runs)
`band-gap`	VBM, CBM and gap of a finished calculation
`vbm-character`	Orbital character of the valence band maximum
`cbm-character`	Orbital character of the conduction band minimum
`band-character`	Orbital character of any band at any k-point
`inspect-mlk`	Show how a Mulliken band file is read (format check)

4.1 execute

Runs the complete correction in a folder (section 7 explains the run in detail).

```
aimshalf execute [--dir DIR] [--config FILE] [--foreground]
                 [--corrections JSON] [--aims-cmd CMD] [--xc CODE]
                 [--cut-min R] [--cut-max R] [--tol R] [--coarse-points N]
                 [--use-band-files]
```

Option	Meaning
`--dir DIR`	calculation folder (default: current folder)
`--config FILE`	YAML file (default: aimshalf.yaml in --dir)
`--foreground`	stay attached to the terminal (automatic inside SLURM/PBS/LSF jobs)
`--corrections JSON`	explicit valence corrections, e.g. `'{"As": {"p": 0.5}}'`
other options	override the yaml keys of the same name (section 10)

Before anything starts, execute checks the inputs and stops with a list of all problems (missing files, aims binary not found, no band lines, unknown species or orbital, invalid correction code, ...).

4.2 gen-sep

Writes one SEP file, as execute does for each run. Useful to look at the potential or to set up a calculation by hand.

```
aimshalf gen-sep SPECIES -c l=dq [-c l=dq ...] --cut R_CUT
                 [--xc pb] [--power 8] [--amplitude 1.0] [--out-dir DIR]

aimshalf gen-sep As -c p=0.5 --cut 3.8
SEP written: sep_As_cut3.800.dat
tail-charge check: fitted q = 0.4383 e (expected ~ 0.5000)
```

- Several `-c` options make several channels (one atomic run each), summed — the scheme execute uses. `--amplitude -1` gives a conduction-type (repulsive) SEP.
- File format: comment lines, number of points, then r (bohr) and V_S (Hartree); exactly zero beyond r_{cut}. In control.in: `aimshalf_potential As sep_As_cut3.800.dat`.

4.3 band-gap

Reads the `band*.out` files of a finished FHI-aims calculation. By default it reports the **smallest direct gap** (as minushalf); `-i` gives the fundamental gap. Example, AlAs (PBE+SOC), which is indirect:

```
aimshalf band-gap --dir pbe_ref
VBM: -0.6691 eV at k-point 1 = (0.0000, 0.0000, 0.0000), state 45
CBM: 1.2794 eV at k-point 1 = (0.0000, 0.0000, 0.0000), state 47
gap: 1.9484 eV (direct)   [smallest direct gap]

aimshalf band-gap --dir pbe_ref -i
VBM: -0.6691 eV at k-point 1 = (0.0000, 0.0000, 0.0000), state 45
CBM: 0.6691 eV at k-point 44 = (0.5000, 0.0000, 0.5000), state 47
gap: 1.3382 eV (indirect)   [fundamental gap (-i)]
```

k-points are numbered 1, 2, ... over all band files in order and states from 1; `band-character` and the `overwrite_*` options use these numbers.

4.4 vbm-character, cbm-character

Mulliken orbital character of the band edge, from the `bandmlk*.out` files (written by FHI-aims for `output band_mulliken`; execute's PBE reference produces them). The weights are summed over all states degenerate with the edge (with SOC the VBM at Γ is 4-fold, and one state of a degenerate set has an arbitrary composition). `-i` uses the edges of the fundamental gap.

```
aimshalf cbm-character --dir pbe_ref          # AlAs: CBM at Gamma (direct gap)
CBM: E = 1.2794 eV at k-point 1 = (0.0000, 0.0000, 0.0000), states [47, 48] summed

by species and orbital:
   Al-s   63.13 %
   As-s   24.22 %
   Al-f   12.01 %

aimshalf cbm-character --dir pbe_ref -i       # AlAs: CBM at X (fundamental gap)
CBM: E = 0.6691 eV at k-point 44 = (0.5000, 0.0000, 0.5000), states [47, 48] summed
   Al-p   48.43 %
   Al-d   30.79 %
   As-d   17.06 %
   As-s    3.66 %
```

> **Note.** Mulliken weights per angular momentum depend on the basis set: with `tight` species the d and f functions of one atom describe tails of its neighbour's orbitals (e.g. the Al-d weight above). Trust the ranking of species, not the exact percentages.

4.5 band-character

```
aimshalf band-character K BAND [--dir DIR]
aimshalf band-character 44 47 --dir pbe_ref    # lowest conduction state at X
```

Same output as the edge commands, for state BAND at k-point K (numbers from `band-gap`).

4.6 inspect-mlk

```
aimshalf inspect-mlk pbe_ref/bandmlk1001.out
```

Prints the number of k-points and states, whether the file has a spin column (SOC or spin-polarized) and one parsed line, to compare with the file when a new FHI-aims version is used.

5. Which gap is maximized

The choice matters for indirect semiconductors. aimshalf follows minushalf:

Setting	Gap maximized	Band-edge characters taken at
`indirect: false` (default)	smallest **direct** gap	the k-point of that direct gap
`indirect: true`	**fundamental** gap (VBM and CBM at any k)	the VBM and CBM k-points

For AlAs (PBE+SOC) the smallest direct gap is at Γ (1.95 eV, CBM Al-s/As-s) and the fundamental gap is Γ→X (1.34 eV, CBM Al-p/Al-d). The two settings therefore select different conduction orbitals and give different r_{cut}. For direct-gap materials (GaAs) they coincide.

> **Important.** Version 0.3.0 changed the default to `indirect: false` (minushalf's). To reproduce results obtained with 0.2, add `indirect: true`.

Choosing the band edges by hand

Key	Effect
`overwrite_vbm: [k, state]`	use this band as the VBM for the character analysis
`overwrite_cbm: [k, state]`	use this band as the CBM
`vbm_characters: [[As, p, 80]]`	replace character values (percent) before the selection
`cbm_characters: [[Ga, s, 70]]`	same for the CBM

6. Choosing the corrections

6.1 Automatic mode (default)

execute first runs plain PBE with Mulliken-resolved bands in `pbe_ref/`, reads the characters of the VBM and CBM (section 5) and applies the **correction code**:

Code	Valence (VBM)	Conduction (CBM)
`v`	dominant orbital	—
`vf`	every orbital above `valence_threshold` (10 %)	—
`c`	—	dominant orbital
`cf`	—	every orbital above `conduction_threshold` (9 %)
`vc`	dominant	dominant
`vfc`	above threshold	dominant
`vcf`	dominant	above threshold
`vfcf`	above threshold	above threshold

- **How much charge.** A single orbital gets the full half electron (dq = 0.5 e). With the fractional codes, each selected orbital gets round(100·p/Σp) percent of it, where p is its character — exactly minushalf's rule. Example: As-p 70 %, Al-p 20 % → 78 % and 22 % → dq = 0.39 and 0.11 e.
- **Empty orbitals** of the free atom (e.g. Al-d) cannot lose charge: they are dropped with a message and the others renormalized.
- The selection is written to `pbe_ref/vbm_character.txt` and to the log.

6.2 Conduction corrections

A `c` in the code adds channels chosen from the CBM character. The atomic calculation is the same as for the valence, but the SEP enters with amplitude −|A|: it is repulsive and raises the conduction band. The valence channels are optimized first and kept fixed; the conduction channels follow, again maximizing the gap.

> **Important.** Conduction corrections can overshoot. For GaAs (light species, no SOC), `v` gives 1.60 eV and `vc` 2.34 eV, against 1.52 eV in experiment. Check against a reference before adopting them.

6.3 Explicit mode

Give the channels and charges yourself; no PBE reference run is made:

```
corrections: {As: {p: 0.5}}              # valence
conduction_corrections: {Ga: {s: 0.5}}   # optional, conduction (needs `corrections`)
```

6.4 Shared holes: divide_character

If an atom's nearest neighbours include atoms of its own species (covalent solids such as Si), the hole is shared and the atom gets 1/(1+n) of its share, n being the number of such neighbours — minushalf's rule (Si: 1/4 e). aimshalf counts n from geometry.in (first neighbour shell, periodic images included). For III-V compounds n = 0. Force the value with `divide_character: [[Si, p, 1]]`.

As in minushalf, n counts distinct atoms in the cell, so it can differ between a primitive and a conventional cell; use divide_character for non-primitive cells.

7. Running execute

7.1 How r_cut is optimized

For each channel: a coarse scan of `coarse_points` cuts between `cut_min` and `cut_max` brackets the maximum, then a golden-section search refines it. One FHI-aims SCF is run per new cut. It stops when the bracket is narrower than `tol` (0.05 bohr), when the top of gap(r_{cut}) is flat within `gap_tol` (5 meV), or after `max_iter` steps.

- If the optimum lies at the edge of the range, a **WARNING** is written: widen `cut_min/cut_max` or use `cut_ranges: {N: [1.5, 4.0]}` per species.
- The top of gap(r_{cut}) is flat: r_{cut} is determined to about ±0.05 bohr. Quote it with that precision (3.81 and 3.83 bohr are the same result).
- Cuts are rounded to `cut_resolution` (0.01 bohr); the same cut is never run twice.

7.2 Background, batch jobs and monitoring

- In a terminal, execute detaches and keeps running after you close it (output in `execute.out`). Inside SLURM/PBS/LSF jobs, or with `--foreground`, it stays attached.
- `aimshalf_status.txt`: state, current run, best result so far, host and PID. If *last update* is more than a minute old, execute is no longer running.
- `aimshalf_execute.log`: timestamped log of every run.
- To stop: `kill PID` (the PID is in the status file). Only one execute can run in a folder at a time.

7.3 Restarting and reusing runs

Run `aimshalf execute` again in the same folder to resume: every finished run is reused **only if it was made from exactly the same inputs**. Each run folder stores `aimshalf_fingerprint.json` (hashes of control.in, geometry.in, the SEP files and the aims binary). If anything changed — another aims version, another correction, another k-grid — the old folder is renamed `stale_<name>_<time>/` (never deleted) and the run is repeated. Changing only `indirect` reuses the runs and recomputes the gaps.

> **Note.** Runs made before version 0.2 have no fingerprint and are redone. To reuse them anyway, set `reuse_unverified: true`.

8. Output

File / folder	Content
`aimshalf_status.txt`	one-glance state and the final result
`aimshalf_execute.log`	timestamped log: selection, every run, optimum, result
`aimshalf_results.dat`	r_{cut}, gap, wall time and state of every run
`pbe_ref/`	PBE reference with Mulliken bands; `vbm_character.txt`
`cut_X.XXX/`	one FHI-aims run per r_{cut} (single channel)
`stepN_<species>-<l>[-cond]/`	one folder per channel when there are several (`-cond`: conduction), each with its `cut_X.XXX/`
`stale_*/`	runs set aside because their inputs changed
`execute.out`	terminal output of a background execute

Example: GaAs, `correction_code: vc` (FHI-aims 261002, light species, no SOC)

```
VBM: E = -0.2686 eV at k-point 1 = (0.0000, 0.0000, 0.0000), 3 degenerate state(s)
     As-p   76.71 %
CBM: E = 0.2686 eV at k-point 1 = (0.0000, 0.0000, 0.0000), 1 degenerate state(s)
     Ga-s   69.88 %
selected channels: As-p (dq = 0.5000 e), Ga-s-cond (dq = 0.5000 e)
--- step 1/2: As-p, dq = 0.5000 e, r_cut in [1.5, 5.0] bohr
As-p: optimal r_cut = 3.820 bohr, gap = 1.5967 eV
--- step 2/2: Ga-s-cond, dq = 0.5000 e, r_cut in [1.5, 5.0] bohr; fixed: As-p @ 3.820
Ga-s-cond: optimal r_cut = 3.040 bohr, gap = 2.3386 eV
RESULT: As-p (dq 0.500) r_cut = 3.820  +  Ga-s-cond (dq 0.500) r_cut = 3.040
        ->  gap = 2.3386 eV
```

Plotting scripts

- `scripts/compare_cuts.py`: copy it into the execute folder and run it with no arguments; it plots gap vs r_{cut}, the conduction minima at Γ, X, L vs r_{cut}, and the band structures of all cuts.
- `scripts/aims_bands_cut.py`: band structure of one calculation (VBM at 0 eV), with a gnuplot script.

9. Worked examples

9.1 Direct-gap III-V, valence only (GaAs)

```
aims_cmd: "mpirun -np 8 /path/aims.x"
correction_code: v
```

The VBM is As-p; one channel; result above (step 1): 0.54 → 1.60 eV, $r_{cut} \approx 3.8$ bohr.

9.2 Indirect semiconductor (AlAs): fundamental gap

```
aims_cmd: "mpirun -np 8 /path/aims.x"
correction_code: v
indirect: true
```

Maximizes Γ→X instead of the direct gap at Γ (section 5). Run both settings if you need to compare with minushalf results, which use `indirect: false` unless set otherwise.

9.3 Fractional valence with conduction (vfcf)

```
correction_code: vfcf
valence_threshold: 10
conduction_threshold: 9
```

Every orbital above 10 % of the VBM and above 9 % of the CBM becomes a channel; valence channels first, then conduction ones; one r_{cut} per channel.

9.4 Explicit correction without the PBE reference

```
corrections: {N: {p: 0.5}}
cut_ranges: {N: [1.5, 4.0]}
```

9.5 Covalent solid (Si)

```
correction_code: v      # Si-p; n = 1 equal neighbour -> dq = 0.25 e
```

The log shows `Si-p: 1 nearest neighbour(s) of the same species -> 1/2 of its share`.

10. Configuration reference (aimshalf.yaml)

Key	Default	Meaning
`aims_cmd`	`mpirun -np 4 aims.x`	command that runs FHI-aims built with the aimshalf hooks
`xc`	`pb`	functional of the atomic calculations (pb = PBE, ca = LDA); must match control.in
`corrections`	auto	auto, or explicit valence channels `{As: {p: 0.5}}`
`conduction_corrections`	none	explicit conduction channels (with explicit corrections)
`correction_code`	`v`	v, vf, c, cf, vc, vfc, vcf, vfcf (section 6)
`valence_threshold`	10.0	% for vf
`conduction_threshold`	9.0	% for cf
`indirect`	false	false: smallest direct gap; true: fundamental gap
`overwrite_vbm` / `overwrite_cbm`	none	[k, state] of the band edge
`vbm_characters` / `cbm_characters`	none	[[species, l, percent], ...] replacements
`divide_character`	from geometry.in	[[species, l, n], ...] equal neighbours
`pbe_ref_dir`	`pbe_ref`	folder of the PBE reference run
`cut_min` / `cut_max`	1.5 / 5.0	r_{cut} search range (bohr)
`cut_ranges`	none	per species range, `{N: [1.5, 4.0]}`
`coarse_points`	5	cuts of the coarse scan
`tol`	0.05	stop when the bracket is narrower (bohr)
`gap_tol`	0.005	stop when the top is flat within this (eV)
`max_iter`	12	golden-section steps after the coarse scan
`cut_resolution`	0.01	cuts are rounded to this (bohr)
`use_band_files`	true	gap from band*.out (needed for indirect: false)
`power`	8	exponent of the trimming function
`amplitude`	1.0	A, scales the SEP (conduction channels use −|A|)
`run_timeout`	0	seconds per aims run, 0 = no limit
`finish_grace`	30	s; stop an MPI launcher still alive after aims finished
`reuse_unverified`	false	reuse pre-0.2 runs without fingerprint

11. Good practice and limitations

- Use structures from plain DFT or experiment; DFT-1/2 total energies and forces are not meaningful.
- Keep `xc` consistent with control.in (PBE ↔ pb).
- Converge the plain-PBE settings first (basis, k-grid); the same settings are used for every cut.

- Compare the corrected gap with experiment or a hybrid/GW reference per valley (Γ, X, L), not only the fundamental gap.
- The atomic potentials are non-relativistic (ATOM) while FHI-aims uses atomic ZORA (+SOC). The SEP is a difference of potentials, which cancels much of this, but check heavy elements (In and beyond).
- Not tested: GPU builds of FHI-aims, spin-polarized systems.
- Not implemented: shell DFT-1/2, occupation-dependent DFT-1/2.

Differences from minushalf

Topic	minushalf (VASP)	aimshalf (FHI-aims)
SEP injection	added to the POTCAR in reciprocal space	added to the real-space Kohn–Sham potential (aimshalf hooks); identical potential, verified
r_{cut} search	Nelder–Mead from an initial guess	coarse scan + golden-section
Degenerate band edges	one state	summed over the degenerate states
Empty orbitals in selection	run fails	dropped with a message
Restarts	recomputed	reused when the fingerprint matches

12. References

- L. G. Ferreira, M. Marques, L. K. Teles, *Phys. Rev. B* **78**, 125116 (2008).
- L. G. Ferreira, M. Marques, L. K. Teles, *AIP Advances* **1**, 032119 (2011).
- minushalf — GMSN group, ITA (minushalf.readthedocs.io): workflow, atomic program and inputs.
- V. Blum *et al.*, *Comput. Phys. Commun.* **180**, 2175 (2009) — FHI-aims.

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